

Geocomputation & Machine Learning course

Quick recap

This was a GIS and remote sensing class session focused on PK Tools and multi-core processing techniques. Professor Giuseppe led a detailed discussion on using PK Tools for raster processing, including PKGetMask for creating thresholds and masks from pixel values, PKSetMask for applying masks to multiple images, and PKComposite for creating temporal composites with masking capabilities. The class covered practical applications like creating water masks, handling quality control flags, and performing image filtering and resampling operations. Students asked questions about handling large vector files (500MB), downloading MODIS and Sentinel data using command-line tools like wget, and implementing multi-core operations using XARG. The session also covered PKStat for generating histograms and statistical analysis, PKFilter for moving window analysis, and PKExtract for zonal statistics. Students shared their specific use cases including flood mapping, land use classification, and satellite time series analysis, with Professor Giuseppe providing guidance on appropriate tools and techniques for their projects.

Next steps

Abdul

- Use PK histogram and reclassification to create a lookup method for scaling water depth data for different return periods, and prepare an example to show to the class.

Annabell

- Simplify the large vector shapefile using OGR2OGR's simplify function for visualization purposes.

Eva

- Resolve authentication issues with Google Earth Engine Python API and consider using Python interface for processing.

Johannes

- Use PKTools to mask clouds in Sentinel-2 time series using the cloud band, and then compute median or mean NDVI for the period.

Shiva

- Download MODIS and Copernicus raster datasets for Italy using wget or curl, and automate the process with a bash script.

Collaboration

- All attendees: Explore the GDAL vs. PKTools comparison table provided by Giuseppe for reference.
- All attendees: Watch the video lecture on multi-temporal interpolation of Landsat images for a practical example of using PKTools and multi-core processing.

Summary

Large Vector File Solutions

Giuseppe discussed solutions for handling large vector files, addressing Annabell's question about a 500MB shapefile that was struggling to load on her laptop. He explained two main approaches: using OGR's simplify function to reduce vertex density based on a tolerance threshold, or converting the vector to raster format using GDAL's rasterize function. Giuseppe recommended using GeoPackage or JSON formats instead of shapefiles and provided guidance on setting appropriate pixel size and extent for rasterization, suggesting a target resolution matching the input raster while maintaining alignment with the vector data.

Vector Data and Raster Processing

Giuseppe provided guidance on simplifying vector data and using GDAL commands for downloading raster datasets. He advised Shiva on using `wget` or `curl` commands for downloading MODIS and Copernicus data, emphasizing the importance of building procedural approaches rather than manual downloads. The discussion also covered the use of Google Earth Engine, with Giuseppe recommending Python over JavaScript interfaces and noting that while Google Earth Engine can be convenient, downloading data directly from original sources provides better control over the processing. Giuseppe concluded by warning about multi-core processing limitations, particularly that multiple cores cannot write to the same file simultaneously, and discussed managing GDAL RAM usage through the `GDAL_MAX` option.

GDAL and PK Tools Comparison

Giuseppe explained GDAL settings and multi-core operations, emphasizing the importance of exporting variables when transitioning between different programming environments like Bash and XARC. He introduced PK Tools, highlighting its historical development and integration with GDAL, and presented a comparison table between GDAL and PK Tools commands to help participants identify appropriate tools for their analyses. The session included a discussion of unique features in both GDAL and PK Tools, with Giuseppe encouraging participants to use the comparison table to build processing chains step by step.

PKTools Raster Operations Demonstration

Giuseppe discussed the use of PKTools for raster operations, particularly focusing on thresholding and masking techniques. He explained how `PKGetMask` can create binary maps by transforming pixel values within a specified range, and demonstrated the process using an example image. Giuseppe also covered `PKSetMask`, which allows applying masks to other images, and highlighted the flexibility of these tools for identifying and manipulating specific features in satellite images.

PKGAT Mask Application Techniques

Giuseppe explained how to use the PKGAT mask to create threshold masks and apply them to raster data, demonstrating examples with digital elevation models and satellite images. He showed how masks can be used to exclude areas like water or poor quality pixels, and explained the process of creating composites by applying masks to stack layers and calculating means or medians for non-masked pixels. The discussion included practical

examples of using quality control flags in satellite images and creating PK composites to analyze data across multiple time periods.

PK Composite Image Processing Methods

Giuseppe explained the PK composite and PK stack profile methods for handling satellite images with different resolutions and quality control bands, emphasizing their flexibility compared to GDAL's composite functionality. He demonstrated how these tools can perform statistical operations like mean and standard deviation calculations across image layers, and discussed various filtering techniques including moving window analysis and resampling methods. The discussion included examples from participants about using these techniques for digital elevation model resampling and temperature data aggregation at different resolutions.

Image Classification and Analysis Techniques

Giuseppe discussed edge detection techniques and their importance for image classification work, particularly for satellite imagery applications. Areeba shared her experience using zonal statistics in ArcGIS Pro with Zoner for land use and land cover mapping of crops. Giuseppe then explained the PK stat tool, which provides statistical information about images including histograms, helping identify pixel distributions and potential errors that might not be visible to the eye.

Histograms in Image Processing Applications

Giuseppe explained the importance and applications of histograms in image processing, particularly using PKStat tools. He demonstrated how histograms can help identify pixel distributions, check masking accuracy, and perform image reclassification using lookup tables. Giuseppe showed examples of using histograms to verify data type conversions, handle integer overflow issues, and reclassify pixel values for land cover analysis. The discussion included practical applications like checking masking steps and performing zonal statistics, with Abdul expressing interest in using histograms for his return period analysis work.

GDAL and PK Tools Demonstration

Giuseppe demonstrated how to use GDAL and PK tools for raster data extraction and processing, including handling multiple files and creating VRT stacks. Abdul plans to use these

tools to create a lookup method for his database analysis, while Sun inquired about averaging data across 12 months, which Giuseppe showed could be done using GDAL with a VRT approach. Johannes requested guidance on masking clouds from Sentinel-2 time series data and creating median values, which Giuseppe suggested could be done using the built-in cloud identification bands and PK composite tools.